

Namma

**AI-Powered Traffic Intelligence Platform
for Incident Prediction, Resource Allocation &
Diversion Planning**

Smarter Roads. Safer Journeys. Faster Decisions.



Flipkart Gridlock 2.0

ML Baddies

Recent Situation

- Bengaluru serves 2.4+ million daily commuters
- Thousands of traffic incidents occur every day
- Frequent congestion hotspots across major roads
- Regular road closures due to construction and maintenance
- Traffic disruptions caused by VIP movements
- Large public events create sudden traffic surges



Scope

- Citizen incident reporting
- Real-time traffic monitoring
- AI-powered priority prediction
- Resource allocation recommendations
- Diversion planning
- Safe route guidance
- Traffic command center analytics

Why Current Systems Fail

Fragmented Reporting Channels

- No single platform provides a unified city-wide traffic view.
- Critical information is often delayed or missed.

Reactive Incident Response

- Authorities typically respond after congestion has already formed.
- Preventive action is rarely possible.

Inefficient Resource Allocation

- Traffic personnel are deployed based on experience rather than real-time data.

No Diversion Planning

- Alternative routes are created manually and reactively.
- Lack of city-wide optimization leads to network-wide bottlenecks.

Limited Citizen Visibility

- Commuters often discover roadblocks only after entering affected routes.
- Existing navigation tools may not capture local incidents quickly enough.
- Citizens lack reliable, safety-focused route recommendations.

Namma Platform

Namma combines citizen reports, machine learning, and diversion planning into a unified traffic intelligence platform that predicts incidents, prioritises response, and reduces congestion before it escalates.

It is available for clients/users, traffic personnel, and authorities. It gives an ultimate control dashboard where authorities can take real-time decisions and regulate event management.

Citizen Report → AI Prediction → Officer Action → Diversion Planning → Safer Roads

What Makes Us Different

Leakage-Free CatBoost Prediction Pipeline

- Road Closure Prediction: **ROC-AUC 0.8095**
- Incident Priority Prediction: **ROC-AUC 0.8155**
- Production-realistic evaluation with Geohash Holdout Testing

Explainable SHAP

- Every prediction includes feature-level explanations
- Displays top contributing factors and impact direction

Tactical Advisories

- Actionable recommendations for field officers

Real-time multilingual translation

- Supports English, Hindi, and Kannada

ASTRAM Diversion Engine

- Automated diversion planning on real road networks
- Built using OSMnx and incident-weighted A* routing
- Generates optimized alternate routes during disruptions

Incident-Aware Safe Routing

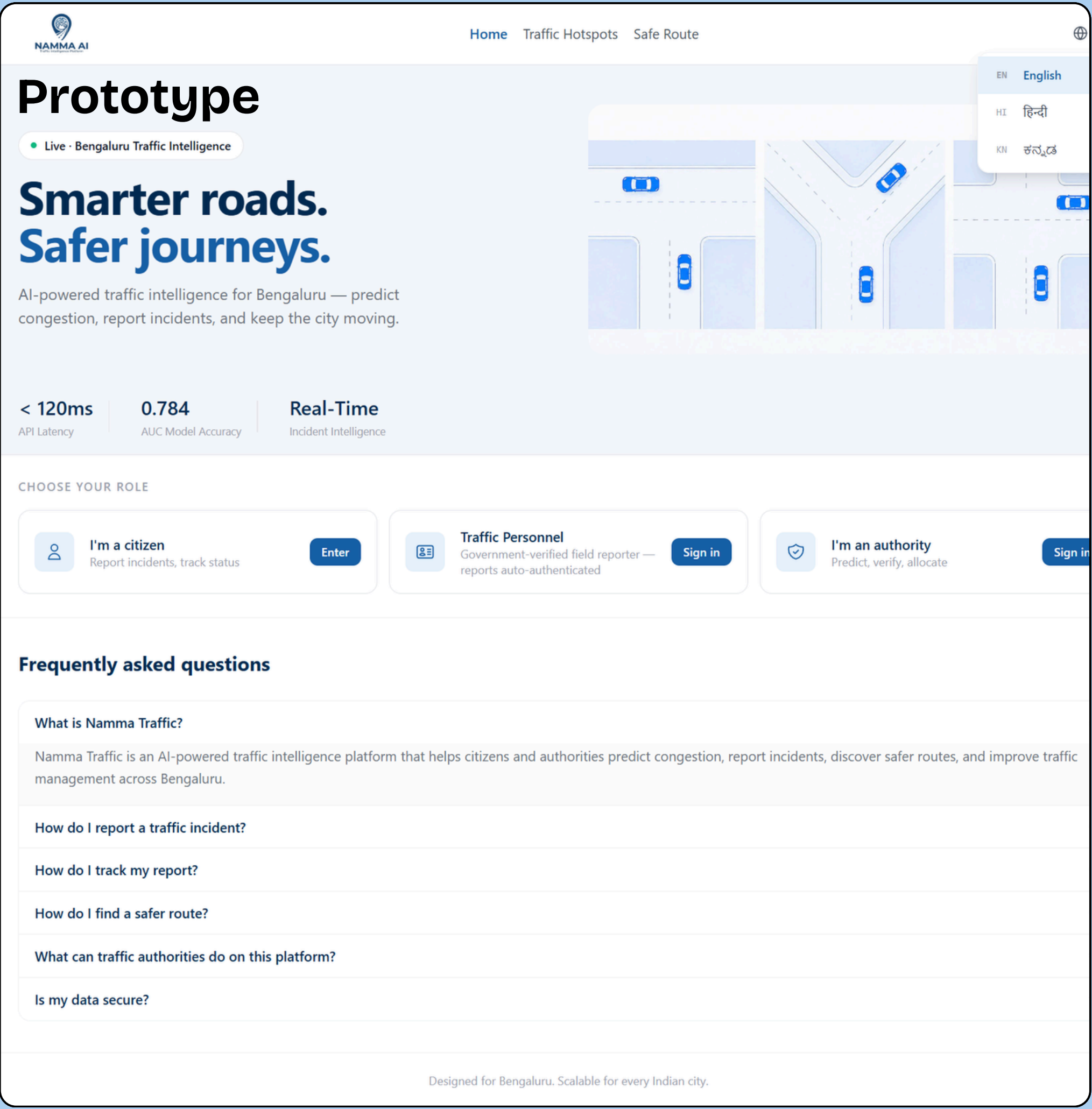
- Live incident severity integrated into route scoring
- Closed roads automatically excluded from navigation
- Safer and more reliable routing for commuters

Real-Time Hotspot Detection

- DBSCAN-based spatial clustering
- Detects emerging traffic hotspots dynamically
- Runs directly on live PostgreSQL incident data

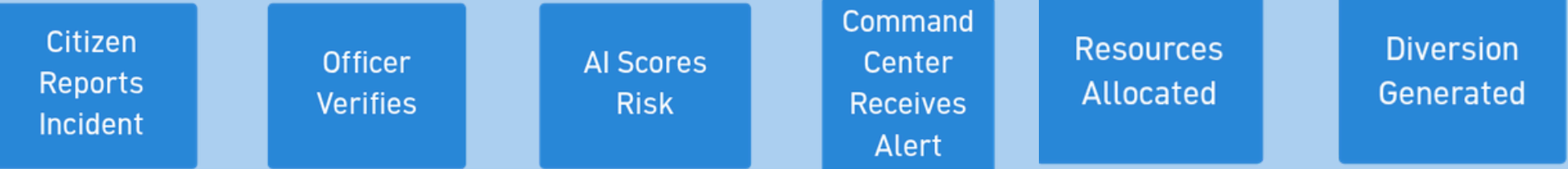
Fault-Tolerant Architecture

- No single-point dependency
- Rule-based fallback mechanisms for all critical features
- Graceful degradation during service outages

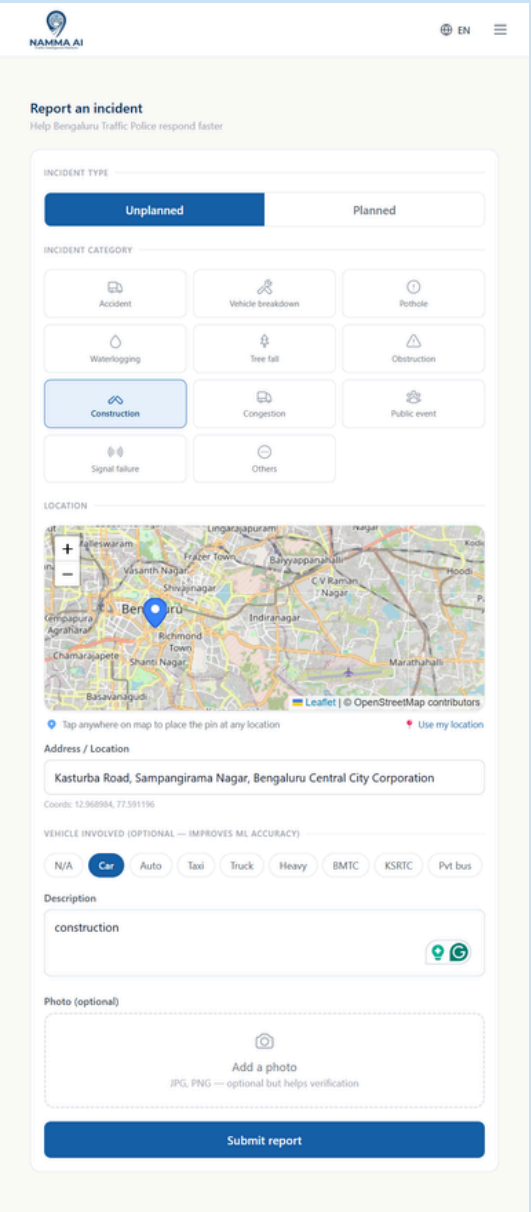


Purpose

A unified, real-time traffic intelligence platform that connects **citizens, traffic officers, and command-centre** operators through a single verified data pipeline, enabling faster incident response and smarter traffic management.

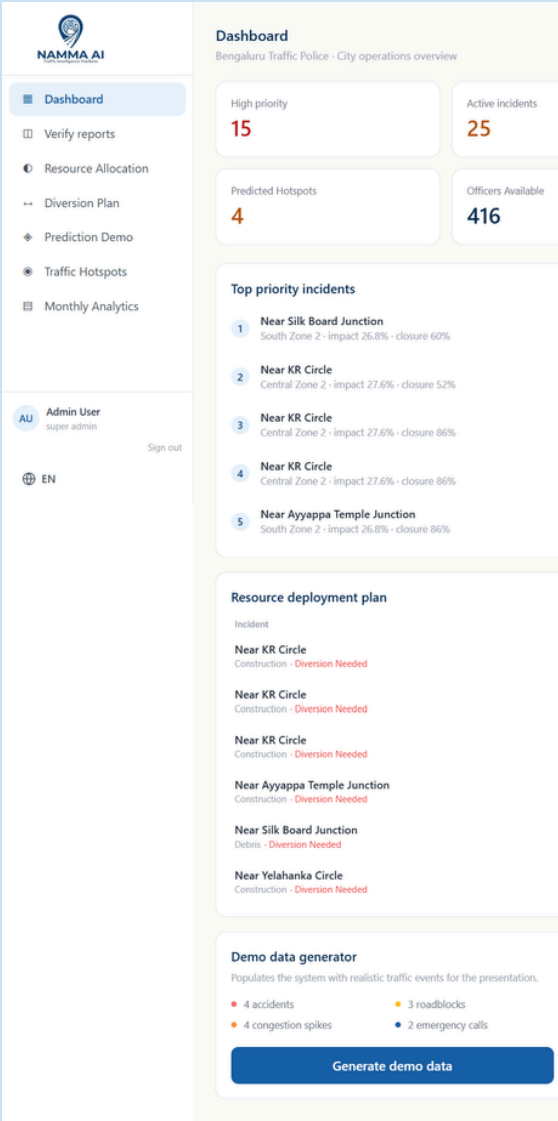


Client/User



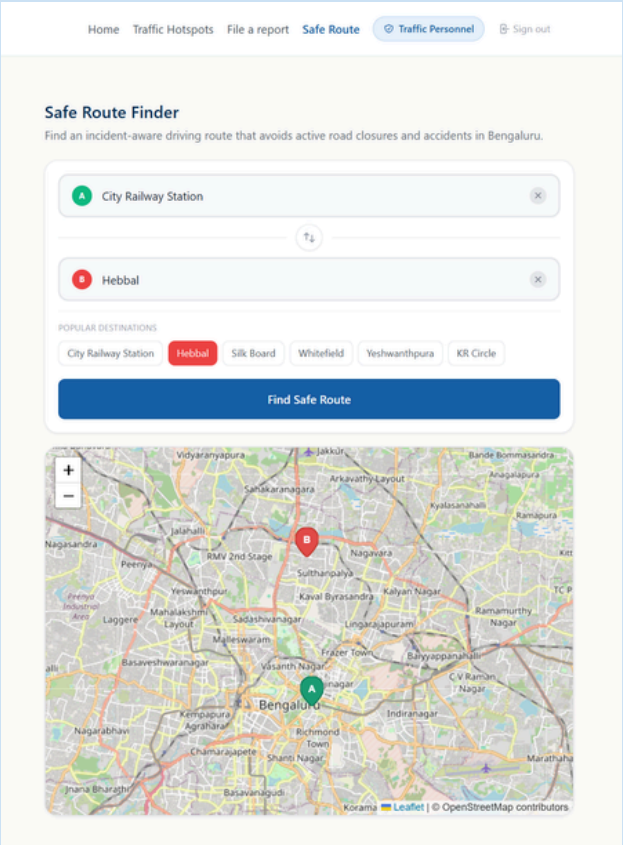
- One-tap incident reporting form
- Live GPS location & photo upload
- Personalized tracking dashboard for all reports
- Receive alerts & diversion plans for relevant areas

Traffic Personnel



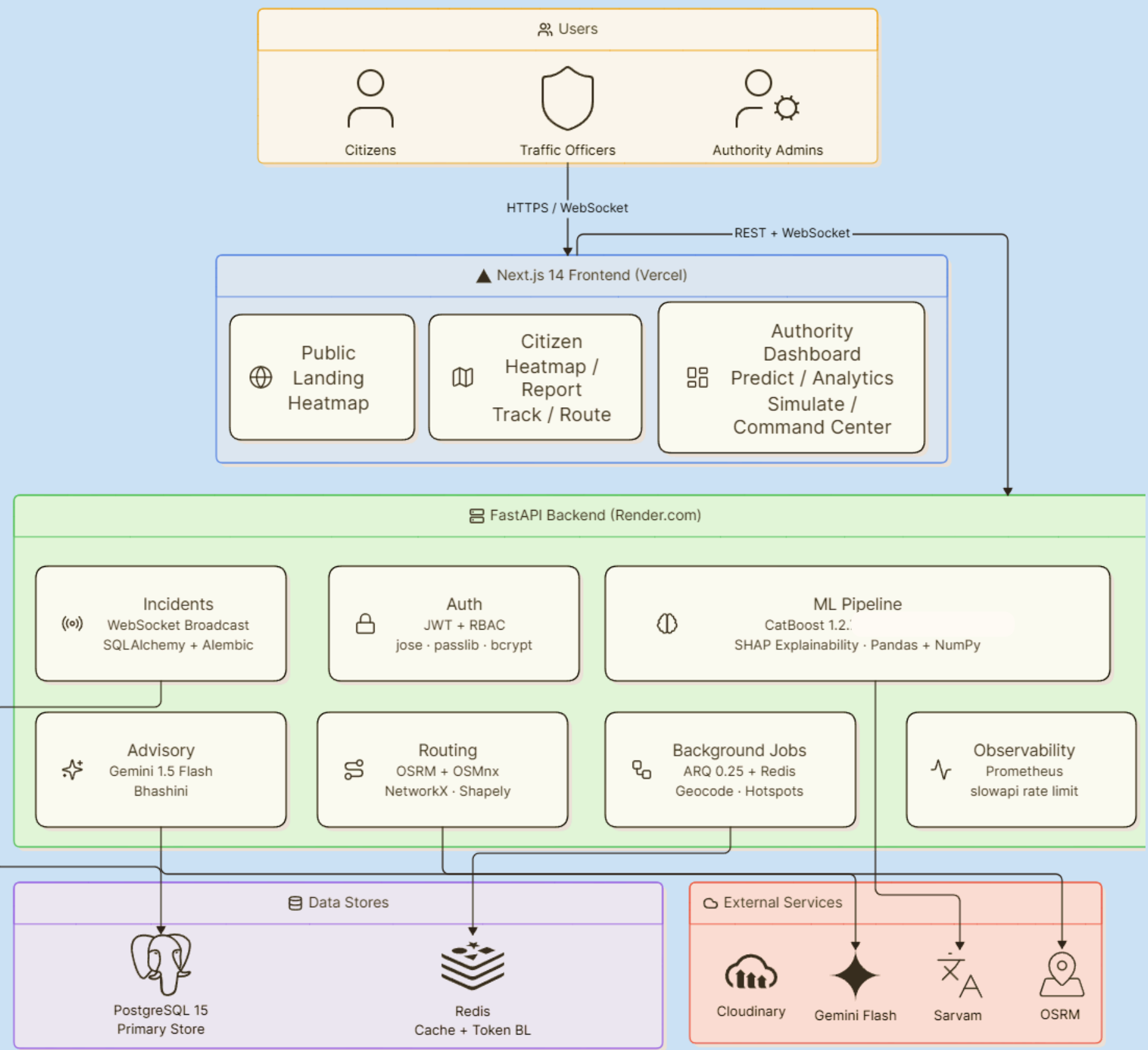
- Multi-level queue for report processing
- Direct action interface (Approved directly as it is by an authenticated authority)
- Access to multi-level role-based controls (RBAC)
- Secure, auditable workflow tracking

Administrator Panel



- Interactive city-wide incident map
- WebSocket-powered live updates
- Instant synchronization across dashboards
- Closure and priority predictions
- SHAP-based feature explanations

Workflow Diagram



Citizen → AI Prediction → Command Center → Diversion Planning → Safe Routing

Layer	Technology Used	Purpose
Frontend	Next.js 14, React 18, TypeScript	Interactive citizen, officer, and admin
Maps & GIS	OpenStreetMap, Leaflet	Traffic heatmaps, incident visualization,
Database	PostgreSQL, SQLAlchemy, Alembic	Storage of incidents, users, reports, and
Authentication	JWT, bcrypt, Passlib	Secure authentication and role-based access
Background Processing	ARQ, Redis	Geocoding, hotspot recalculation, asynchronous jobs
Diversion Planning	OSMnx, NetworkX, OSRM	Road graph creation and intelligent
Backend	FastAPI, Uvicorn	High-performance REST APIs,
Monitoring	Prometheus	API latency, uptime, and system
API Communication	REST APIs, WebSockets	Real-time frontend-backend
Deployment	Vercel, Render, Docker	Cloud hosting, CI/CD, and application

Key Insights from ML

Geography Dominates Traffic Outcomes

- Our experiments revealed that spatial information is the **strongest** predictor of **incident severity** and **operational impact**.
- **Location-based features** contributed 62.92% of total model importance, significantly outperforming temporal and categorical attributes.
- **Interpretation:** Traffic behavior is highly localized. Certain road segments consistently exhibit **higher risk profiles** due to **infrastructure constraints, traffic volume, and historical disruption patterns**.

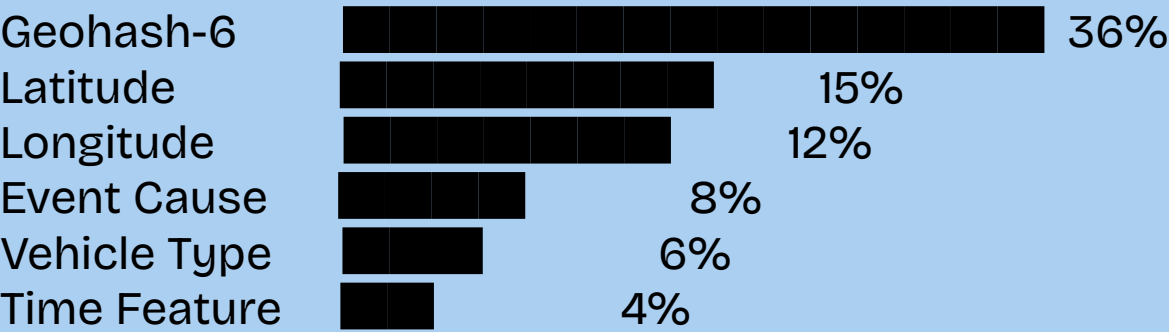
Historical Spatial Patterns Remain Valuable

- Even after removing target leakage, historical geographic behaviour continued to provide meaningful predictive power.
- **Examples:** • Regions with **frequent** incidents remained **higher-risk zones**
- • **Historical closure rates** improved **closure prediction accuracy**
- • **Spatial incident density** helped identify recurring **hotspots**
- **Interpretation:** Traffic incidents are not randomly distributed. Geographic history contains actionable operational intelligence.

Closure Risk Improves Priority Prediction

- The **Closure Prediction Model** was used as an upstream signal for **Priority Prediction**.
- **Out-of-fold closure** probabilities were incorporated into the priority model.
- **Result:** • Closure probability became an informative secondary feature
- • Improved decision consistency between models
- • Enabled multi-stage risk assessment
- **Interpretation:** Incidents likely to cause road closures are also more likely to require higher operational priority.

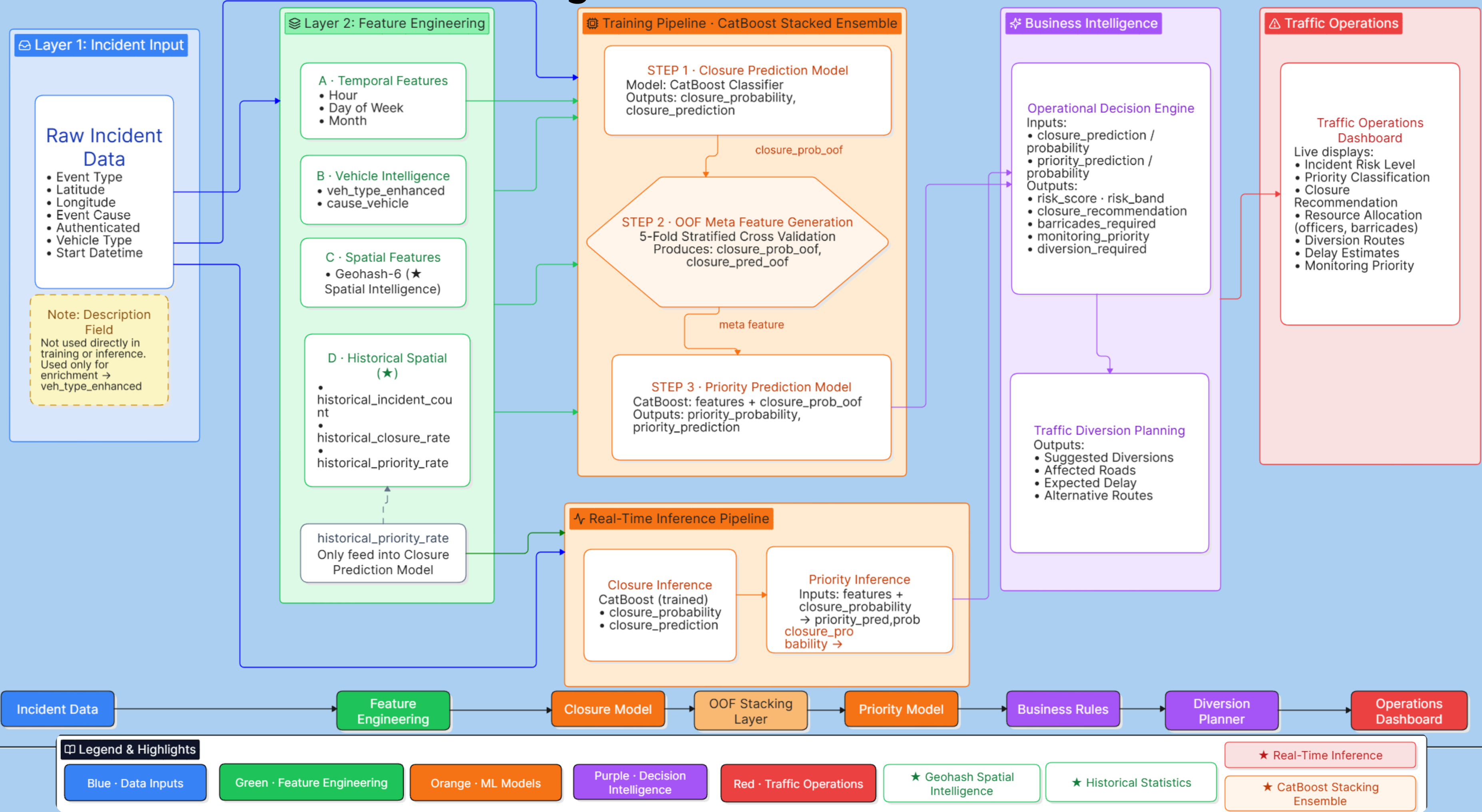
Location-based features account for nearly two-thirds of predictive power.



Structured Data Outperformed NLP Features

- Multiple experiments were conducted using incident descriptions and NLP-derived features.
- **Observations:** • Descriptions were generally short and repetitive
- • Most operational information already existed in structured fields
- • Transformer embeddings produced negligible performance gains
- • CatBoost naturally captured relationships from structured attributes
- **Final Decision:** NLP is used only as a fallback mechanism when critical structured fields are missing or unknown.
- **Examples:** • Missing Event Cause, Vehicle Type or Authentication Status
- **Interpretation:** For traffic incident management, **high-quality structured data is** more valuable than free-text descriptions.

System Architecture



The Hidden Problem

Many traffic prediction systems report high accuracy due to data leakage, where future information unintentionally becomes available during training.

Affected Features

- Historical Incident Count
- Historical Closure Rate
- Historical Priority Rate

Risk

- Inflated validation scores
- Unrealistic production expectations
- Poor real-world generalization



Model Performance & Validation
Traffic Closure Prediction Model

ROC-AUC: 0.8095
PR-AUC: 0.4733
F1 Score: 0.4643

Incident Priority Prediction Model

ROC-AUC: 0.8155
PR-AUC: 0.8460
F1 Score: 0.8161

Solution

We redesigned the pipeline using **Out-of-Fold Spatial Feature Generation**.

For each fold:

- Spatial statistics were built only from training data
- Validation data received statistics generated exclusively from training folds
- No target information leaked across folds

This replicated real-world deployment conditions.

Validation Framework

- Leakage-Free Spatial Features
- Out-of-Fold (OOF) Feature Generation
- Geohash Holdout Testing
- Production-Realistic Evaluation
- No Future Information Leakage

Key Achievement

Both models maintained ROC-AUC above 0.80 after eliminating data leakage, demonstrating strong predictive performance under realistic deployment conditions.

Business Rules Engine

Converting AI Predictions into Operational Decisions

Risk Score Formula

$$\text{Risk Score} = 0.4 \times \text{Closure Probability} + 0.3 \times \text{Priority} + 0.2 \times \text{Congestion} + 0.1 \times \text{Weather}$$

Risk Classification Matrix

0-29	Low	Monitor
30-54	Medium	Deploy Local Team

Officers Required

$$\text{Officers} = \text{Base} + \text{Severity Factor} + \text{Congestion Factor}$$

Risk Classification Matrix

55-74	High	Resource Escalation
75+	Critical	Immediate Response & Diversion

Resource Allocation Logic

Barricades Required

$$\text{Barricades} = \text{Road Type} \times \text{Road Type Weight} \times \text{Diversion Need}$$

Administrative Dashboard



Dashboard

Verify reports

Resource Allocation

Diversion Plan

Prediction Demo

Traffic Hotspots

Monthly Analytics

AUAdmin User
super admin

Sign out

EN

Dashboard

Bengaluru Traffic Police · City operations overview

Dashboard KPIs

- Active Incidents
- Road Closures
- Predicted Hotspots
- Average Closure Risk
- Available Officers
- Diversions Generated
- Average Response Time

Updated 2:50:35 PM

Live · 15s

High priority6

Active incidents12

Road closures9

Total tracked12

Predicted Hotspots2

Officers Available459

Emergency Routes9

Advisories Issued2

Top priority incidents

ranked by severity · congestion · closure risk

1

Near Ayyappa Temple Junction

South Zone 2 · impact 26.8% · closure 86%

Medium

54.3

2

Near KR Circle

Central Zone 2 · impact 27.6% · closure 79%

Medium

52.5

3

Near Satellite Bus Stand

West Zone 1 · impact 25.9% · closure 56%

Critical

52

4

Near Yelahanka Circle

North Zone 1 · impact 26.2% · closure 46%

Critical

46.9

5

Near Yelahanka Circle

North Zone 1 · impact 26.2% · closure 80%

Medium

41.8

Resource deployment plan

For every incident the system recommends:

- Recommended Officers
- Officers = f(Priority,Congestion,ClosureRisk)
- Recommended Barricades
- Barricades = f(RoadType,DiversionNeed,Severity)

Computed from incident severity and closure risk

Incident

Near Ayyappa Temple Junction

Construction · Diversion Needed

19

12

Incident

Near Yelahanka Circle

Construction · Diversion Needed

18

12

Real-Time Incident Monitoring

- Live traffic incident feed
- Active incident tracking across the city
- Road closure monitoring
- Continuous dashboard refresh (15-second updates)

Priority Ranking

- Automatically ranks incidents by:
 - Severity
 - Congestion impact
 - Closure probability

Diversion Planning Support

- Flags incidents requiring diversions
- Integrates with ASTRAM Diversion Engine
- Generates alternate traffic movement plans

Intelligent Resource Allocation

- Recommends officers required per incident
- Suggests barricade deployment counts
- Based on:
 - Priority score
 - Closure probability
 - Historical congestion patterns

Live Traffic Hotspot Detection

- DBSCAN-based hotspot clustering
- Identifies emerging congestion zones
- Enables proactive deployment before traffic escalates

Dashboard enabling Bengaluru Traffic Police to monitor incidents, prioritise response, allocate resources, and coordinate diversions in real time.

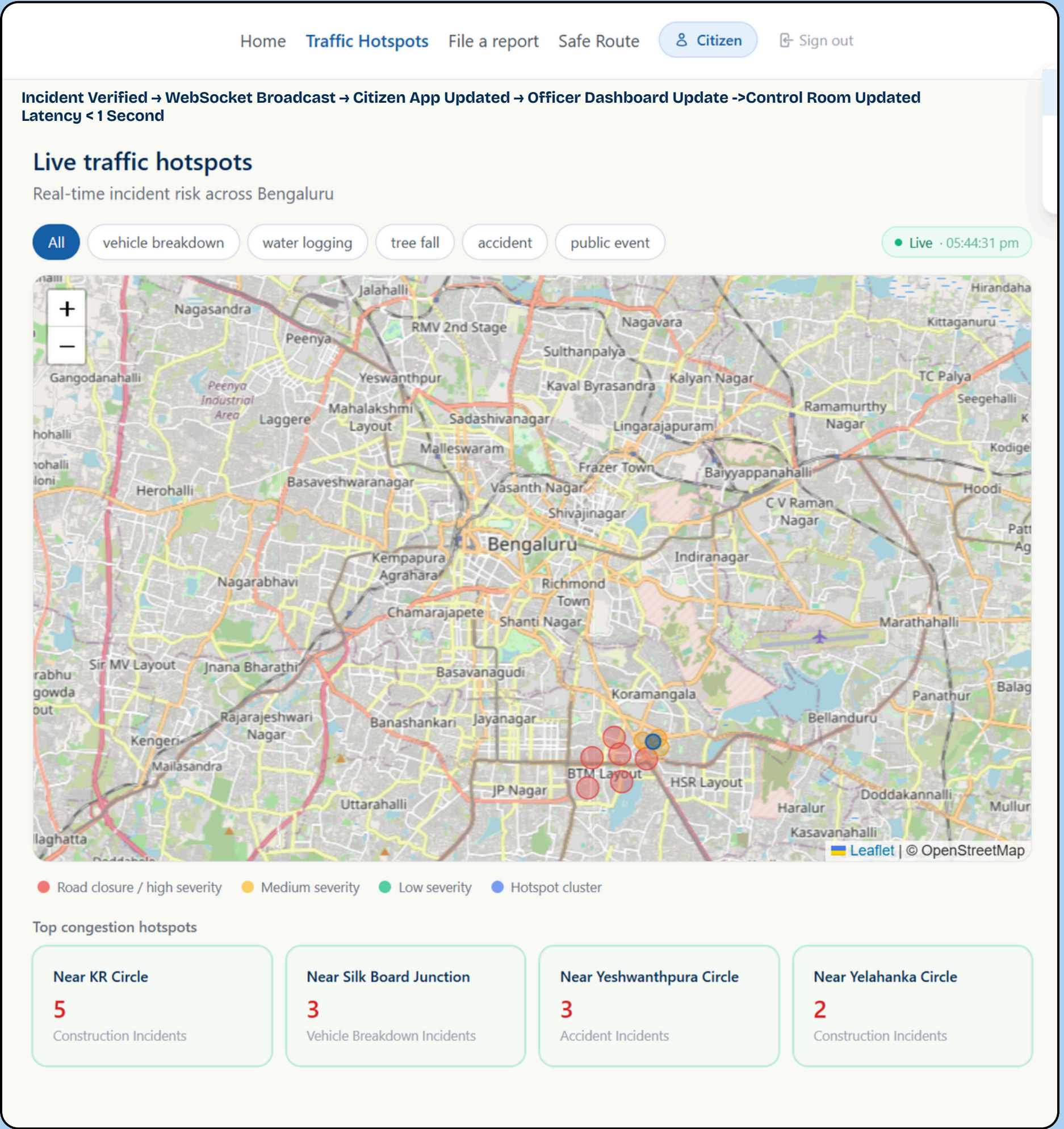
Live Traffic Hotspots Detection

Purpose

- Identifies emerging congestion hotspots from live incident data to support proactive traffic management and resource deployment.
- Supports proactive deployment before congestion escalates into network-wide disruption.

Technical Implementation

- DBSCAN-based spatial clustering
- Haversine geographic distance metric
- 500 m hotspot detection radius
- Minimum 2 incidents per cluster
- Refresh every 15 minutes
- Live PostgreSQL incident stream



I25

Critical

Road closure

Near Silk Board Junction

South Zone 2 · debris · 60% closure risk

Affected road	Road status	Severity	Diversion
Unnamed Road	PARTIALLY BLOCKED	HIGH	Required

Incident 51876b95-4857-4a23-af15-2d18df012e83 (debris) has impacted Unnamed Road. Road status: PARTIALLY_BLOCKED. Severity: HIGH. Diversion required.

Alternative routes

1

Hosur Road

trunk · 23m away

2

5th Cross Road

residential · 38m away

Resource allocation

Recommended deployment · end events to release officers and barricades

Officers deployed

84

Barricades deployed

63

Active incidents

25

Officer availability

0

Per-incident deployment plan

Incident	Location	Officers	Barricades	Radius
I25 Debris Critical	Near Silk Board Junction	2	2	300 m

Affected road: Unnamed Road

PARTIALLY BLOCKED

HIGH severity

Diversion required

Recommended diversions (best first):

1

Hosur Road

trunk

23 m away

2

5th Cross Road

residential

38 m away

Incident 51876b95-4857-4a23-af15-2d18df012e83 (debris) has impacted Unnamed Road. Road status: PARTIALLY_BLOCKED. Severity: HIGH. Diversion recommended.

Diversion Plan

Purpose: Automatically generates optimal diversion routes around incidents and road closures, minimizing congestion spread and maintaining smooth traffic flow across the road network.

Technology Stack

- OpenStreetMap (Road Network Data)
- OSMnx (Road Graph Generation)
- NetworkX (Graph Analysis)
- Incident-Aware A* Routing (Optimal Path Search)

Example Incident: MG Road Closure

Recommended Diversions

- 1.Church Street
- 2.Brigade Road
- 3.Residency Road

Safe Route Scoring

Route Score= Distance+ Traffic Impact+ Incident Risk+ Closure Penalty

Lowest-risk route is recommended.

- Primary Diversion Route
- Alternate Route Options
- Road Closure Bypass Plans
 - Traffic Redistribution Strategy
- Command Center Recommendations

Impact

Traffic Police

- Faster response
- Better resource allocation
- Data-driven decisions

Citizens

- Safer routes
- Faster travel
- Easier reporting

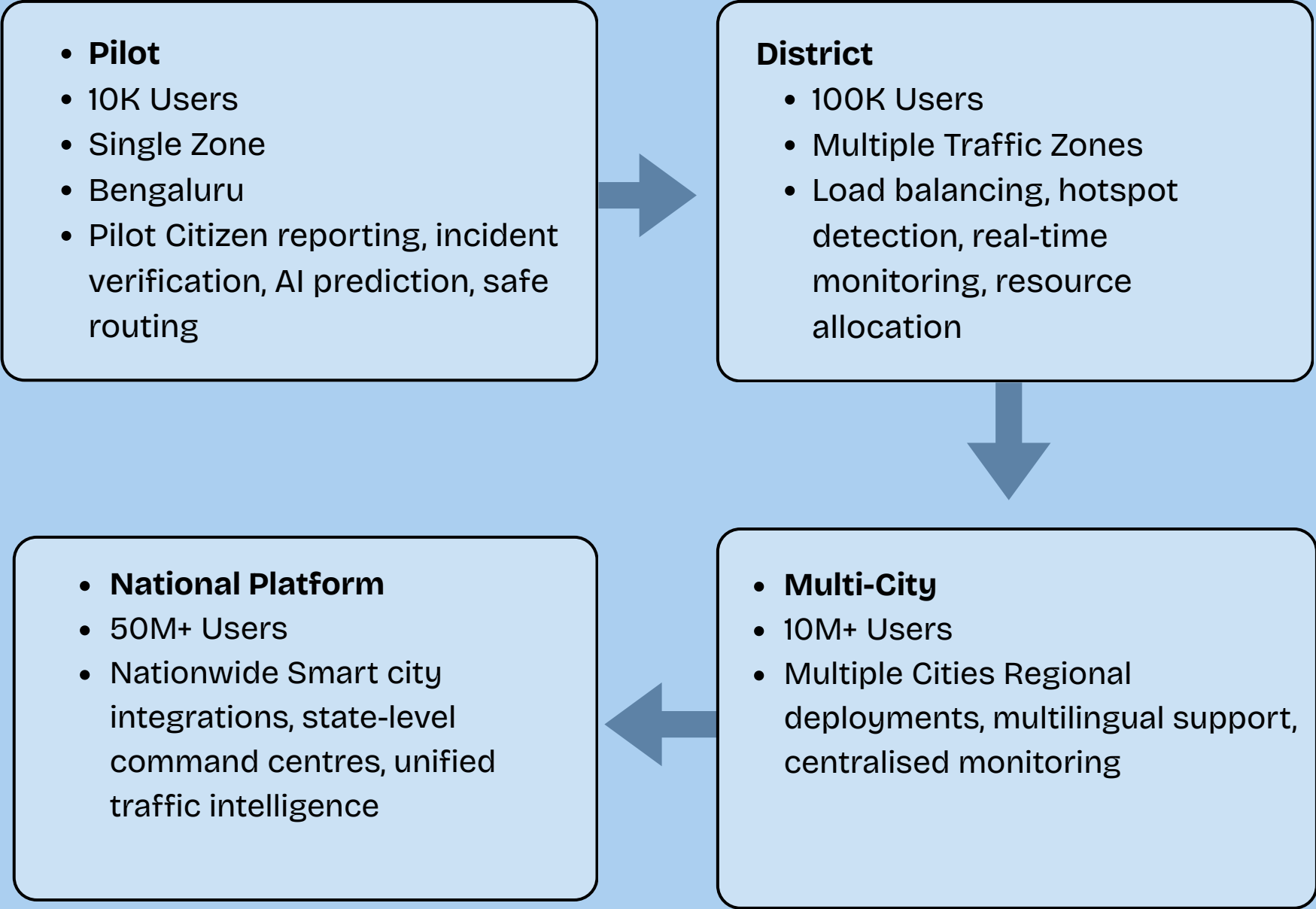
City

- Reduced congestion
- Lower emissions
- Better traffic planning

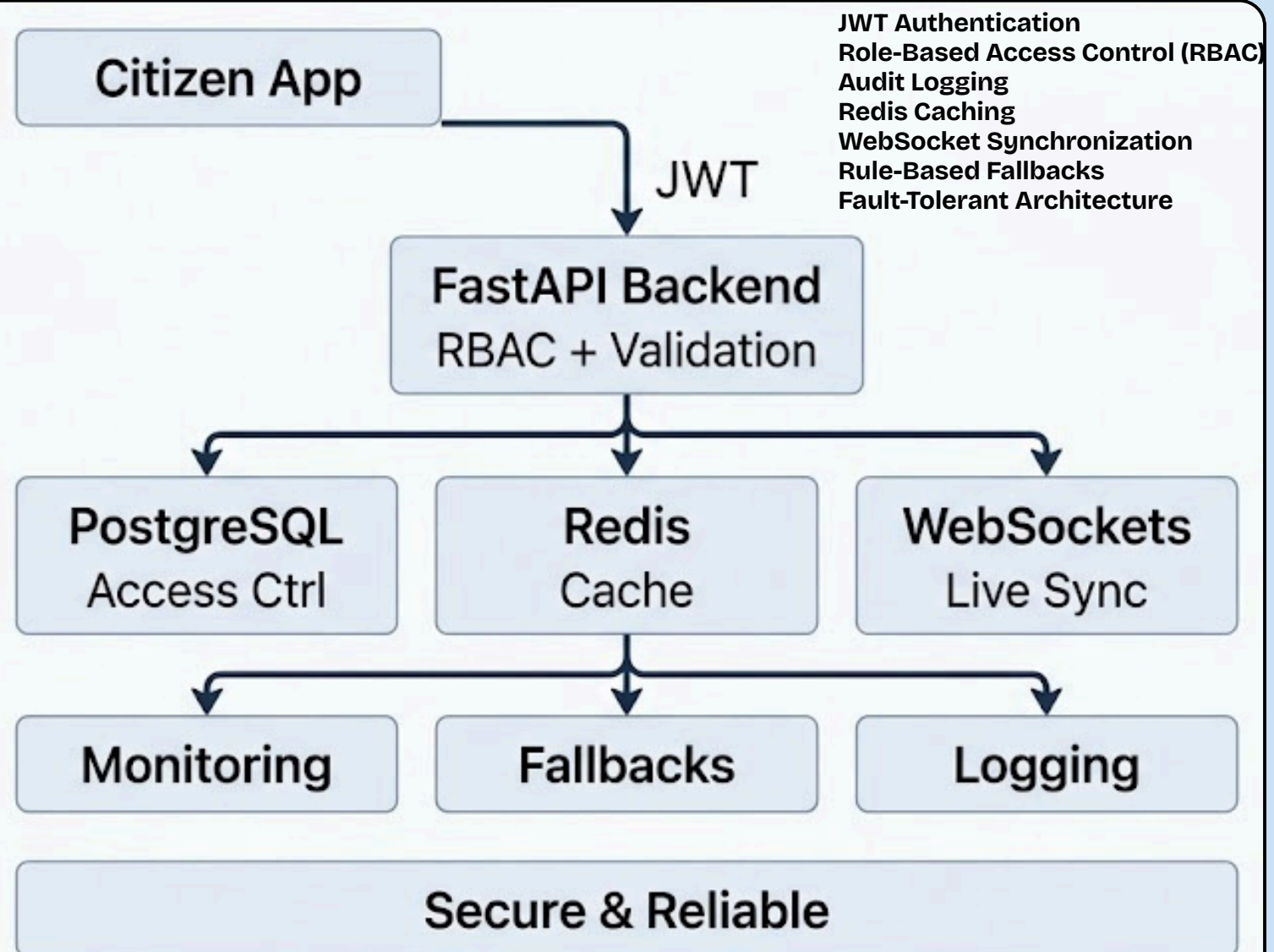
Feature	Google Maps	Waze	Traditional Control Room	Namma Traffic
Citizen Reporting	Limited	YES	No	YES
Officer Dashboard	No	No	Manual	YES
Resource Planning	No	No	Manual	YES
Multilingual Support	YES	NO	NO	YES
Hotspot Detection	NO	NO	NO	YES
Decision Support	NO	NO	NO	YES
Role-Based Access	NO	NO	Limited	YES
Diversion Planning	NO	Limited	Manual	YES
Safe Route Guidance	Basic	Basic	NO	YES
Live Updates	Limited	Partial	NO	YES
Unified Traffic Platform	NO	NO	Partial	YES

Metric	Current	With Namma AI
Incident Verification	Manual	Assisted
Resource Allocation	Experience-based	Data-driven
Diversion Planning	Reactive	Automated
Citizen Alerts	Delayed	Real-time
Hotspot Detection	Manual	DBSCAN-driven

Scalibility



Security



Data Verification Layer

- Role-based verification
- Authority-authenticated reports
- Photo evidence support
- Audit logs for every action
- Multi-stage approval workflow

Economic Feasibility

Component	Technology	Estimated Cost
Frontend Hosting	Vercel	₹0 – ₹500/month
Backend API	Render Cloud Instance	₹2,000 – ₹4,000/month
ML Inference Service	FastAPI + CatBoost	₹2,000 – ₹3,000/month
PostgreSQL Database	Managed PostgreSQL	₹1,000 – ₹2,000/month
Maps & Routing	OpenStreetMap + OSMnx	Free
Monitoring & Storage	Logs, Backups, Analytics	₹1,000/month
IoT Pilot Setup (Future)	10 Cameras + 5 Sensors + 2(RSUs)	₹3 – ₹5 Lakhs (One-Time)

Future Smart-City Expansion:
₹3–₹5 Lakhs (IoT Pilot Infrastructure)

Estimated Operational Cost:
₹5,000–₹10,000/month (Pilot Deployment)

Technical Feasibility

- Built using proven technologies (FastAPI, PostgreSQL, CatBoost, OSMnx, React)
- Modular microservice architecture
- Cloud deployable on Vercel + Render
- Supports real-time incident processing and route planning
- Easily scalable across multiple zones and cities

Operational Feasibility

- Fits existing Traffic Control Room workflows
- Requires minimal operator training
- Citizens can report incidents through a simple interface
- AI recommendations assist rather than replace human decisions
- Supports 24×7 monitoring and incident management

Scalability Feasibility

- Current Scope:
- Bengaluru Pilot Deployment
- Future Scope:
- Multi-Zone Deployment
- Multi-City Deployment
- State-Wide Smart Traffic Network

Sustainability Feasibility

- Expected Benefits:
- Reduced traffic congestion
- Faster incident response
- Lower fuel consumption
- Reduced carbon emissions
- Improved road safety
- Better utilization of traffic personnel

Legal & Data Feasibility

- Uses OpenStreetMap licensed data
- No sensitive personal data collection
- Supports Smart City initiatives
- Compatible with public-sector deployment requirements

Future Roadmap & Innovation Vision

From AI-Assisted Traffic Management to a Self-Learning Intelligent Transportation System

